DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS302: DESIGN AND ANALYSIS OF DEEP NEURAL NETWORKS (4 CREDITS)

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1. COURSE OVERVIEW AND PREREQUISITES

This course provides a rigorous mathematical and practical foundation for modern deep learning architectures. Students will study multi-layer perceptrons, convolutional neural networks (CNNs), recurrent neural architectures (RNNs, LSTMs, GRUs), and transformer-based self-attention mechanisms.

Prerequisites:
- CS201: Data Structures and Algorithms (Minimum Grade: C)
- MA202: Linear Algebra and Multivariate Calculus (Minimum Grade: C)
- Proficiency in Python 3.10+, NumPy, and PyTorch 2.x.

2. DETAILED MODULE BREAKDOWN AND LECTURE SCHEDULE

MODULE 1: FOUNDATIONS OF DEEP LEARNING AND OPTIMIZATION (WEEKS 1–3)

Topic 1.1: Linear Regression and Logistic Regression from scratch using NumPy.

Topic 1.2: Neural Network Topologies:
- Forward Propagation
- Activation Functions:
  - Sigmoid
  - Tanh
  - ReLU
  - LeakyReLU
  - GELU
  - Softmax

Topic 1.3: Vectorized Backpropagation Derivations:
Applying the multivariable chain rule to compute loss gradients with respect to weight matrices and bias vectors.

Topic 1.4: Optimization Algorithms

- Stochastic Gradient Descent (SGD) with Nesterov Momentum.

- Adaptive Gradient Algorithms:
  - AdaGrad
  - RMSProp
  - Adam (Adaptive Moment Estimation)

Adam Parameter Equations:

$$m_t = \beta_1 * m_{(t-1)} + (1 - \beta_1) * g_t$$

$$v_t = \beta_2 * v_{(t-1)} + (1 - \beta_2) * (g_t)^2$$

$$\hat{m}_t = m_t / (1 - \beta_1^t)$$

$$\hat{v}_t = v_t / (1 - \beta_2^t)$$

$$\theta_{(t+1)} = \theta_t - (\eta * \hat{m}_t) / (\sqrt{\hat{v}_t} + \epsilon)$$

MODULE 2: CONVOLUTIONAL NEURAL NETWORKS (CNNs) AND COMPUTER VISION (WEEKS 4–7)

Topic 2.1: Convolution Operation
- Kernel Size
- Stride
- Padding (Valid vs. Same)
- Receptive Fields
- Multi-channel Feature Maps

Topic 2.2: Pooling Layers
- Max Pooling
- Average Pooling
- Global Average Pooling

Topic 2.3: Landmark Architectures
- LeNet-5
- AlexNet
- VGG-16
- ResNet (Residual Connections solving vanishing gradient problems)
- EfficientNet

Topic 2.4: Object Detection and Segmentation Foundations
- YOLO (You Only Look Once) architecture overview
- IoU (Intersection over Union) calculations

MODULE 3: SEQUENCE MODELS AND ATTENTION MECHANISMS (WEEKS 8–11)

Topic 3.1: Recurrent Neural Networks (RNNs)
- Exploding Gradient Problem
- Vanishing Gradient Problem over long sequence time steps

Topic 3.2: Long Short-Term Memory (LSTM) Networks
- Cell State
- Hidden State

- Input Gate
- Forget Gate
- Output Gate Mechanics

Topic 3.3: Gated Recurrent Units (GRUs)
- Reset Gate
- Update Gate
- Trade-offs relative to standard LSTMs

Topic 3.4: Attention Mechanisms
- Bahdanau Additive Attention
- Luong Multiplicative Attention

MODULE 4: TRANSFORMERS AND GENERATIVE MODELS (WEEKS 12–15)

Topic 4.1: Transformer Architecture (Vaswani et al., 2017)
- Scaled Dot-Product Attention
- Multi-Head Attention
- Positional Encodings (Sinusoidal and Absolute)

$$\text{Attention}(Q, K, V) = \text{Softmax}((Q \times K^T) / \sqrt{d\_k}) \times V$$

Topic 4.2: Encoder-Only vs. Decoder-Only Models
- BERT architecture principles
- GPT architecture principles

Topic 4.3: Generative Adversarial Networks (GANs)
- Minimax game objective formulation between Generator (G) and Discriminator (D)

3. GRADING AND EVALUATION POLICY

Continuous Internal Evaluation (CIE): 50%

- Mid-Term Examination: 20%
- Quizzes (3 total, lowest score dropped): 10%
- Laboratory Assignments and Mini-Project: 15%
- Class Participation and Attendance: 5%

End-Term Examination (ETE): 50%

- Cumulative written theory examination covering Modules 1 through 4.

4. TEXTBOOKS AND RECOMMENDED READING

1. Goodfellow, I., Bengio, Y., and Courville, A. (2016). Deep Learning. MIT Press.

2. Zhang, A., Lipton, Z. C., Li, M., and Smola, A. J. (2023). Diving into Deep Learning. Cambridge University Press.

3. Vaswani et al. (2017). "Attention Is All You Need." Advances in Neural Information Processing Systems (NeurIPS).